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COMMENTS ON THE PAPER BY MINXIAN YANG: "SOME PROPERTIES OF VECTOR AUTOREGRESSIVE PROCESSES WITH MARKOV-SWITCHING COEFFICIENTS"

CHRISTIAN FRANCQ
Université du Littoral–Côte d'Opale

JEAN-MICHEL ZAKOÏAN
GREMARS
Université de Lille 3
and
CREST

This paper discusses the stationarity conditions proposed by M. Yang (2000, *Econometric Theory* 16, 23–43), in the framework of Markov-switching first-order autoregressions. A weaker second-order stationarity assumption is proposed.

We were very interested in reading this paper recently published in *Econometric Theory*. We will draw attention to several points because we think that some of the results presented in this paper can be improved. We first briefly recall the notations of the paper. The model is written as a first-order vector autoregression driven by a K-state Markov chain s_t:

$$\mathbf{x}_t = \alpha_{s_t} + \phi_{s_t}\mathbf{x}_{t-1} + \mathbf{u}_t, \tag{1}$$

where $\mathbf{u}_t$ is a noise (whose variance conditional on s_t is λ_{s_t}). The following matrix is introduced:

$$\hat{\mathbf{\Phi}} = \begin{pmatrix} \|\phi_1\| & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \|\phi_K\| \end{pmatrix},$$

where $\mathbf{x}_t$ is an n-dimensional vector and $\|\cdot\|$ is the matrix norm induced by the vectorial 1-norm. The main assumption of the paper is as follows.

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Assumption 2. $\max\{\mathbf{r}(\hat{\mathbf{\Phi}}^2\mathbf{P}), \mathbf{r}(\hat{\mathbf{\Phi}}\mathbf{P})\} < 1$.

In Assumption 2 $\mathbf{P} = (p(j,i))$ is the transpose of the transition probabilities matrix, and $\mathbf{r}(\cdot)$ denotes the spectral radius. The author shows that Assumption 2 is a sufficient condition for the existence of second-order stationary solutions to model (1).

Our remarks do not concern the validity of the main result. In fact we have detected an error in one graph, but this is a minor point. More important, we can prove that part of Assumption 2 is redundant in the univariate case and that, in the multivariate case, Assumption 2 is too strong. We suggest a weaker assumption.

We now detail these three issues.

1. In Section 4 the author considers a two-regime AR(1) model and provides a more explicit expression to Assumption 2. The graph displaying the stationarity region is displayed on page 32. From this figure it seems that the condition $\mathbf{r}(\hat{\mathbf{\Phi}}\mathbf{P}) < 1$ can be more compelling than $\mathbf{r}(\hat{\mathbf{\Phi}}^2\mathbf{P}) < 1$ (in a small region around $\phi_1 = 1$ the two curves intersect at two points). In view of our second point, we were surprised by this, and we have drawn the same graph, which is shown in Figure 1. It is seen that one of the two curves is always below the other one. We would also like to point out that $\mathbf{r}(\hat{\mathbf{\Phi}}^2\mathbf{P}) < 1$ can be simply written as

$$\begin{cases} p(1,1)\phi_1^2 + p(2,2)\phi_2^2 + [1 - p(1,1) - p(2,2)]\phi_1^2\phi_2^2 < 1 \\ p(1,1)\phi_1^2 + p(2,2)\phi_2^2 < 2 \end{cases} \tag{2}$$

and that the other condition is analogous, with squares replaced by absolute values. Figure 1 shows that, in this case, $\mathbf{r}(\hat{\mathbf{\Phi}}^2\mathbf{P}) < 1$ implies $\mathbf{r}(\hat{\mathbf{\Phi}}\mathbf{P}) < 1$.

2. More generally, in the univariate AR(1) case, the condition $\mathbf{r}(\hat{\mathbf{\Phi}}\mathbf{P}) < 1$ is always superfluous. Let $\mathbf{v}_t = \alpha_{s_t} + \mathbf{u}_t$. By iterating equation (1) we obtain

$$\begin{aligned} \mathbf{x}_t &= \mathbf{v}_t + \phi_{s_t}\mathbf{v}_{t-1} + \phi_{s_t}\phi_{s_{t-1}}\mathbf{x}_{t-2} \\ &= \sum_{k=0}^{K} \mathbf{v}_{t,k} + \phi_{s_t}\dots\phi_{s_{t-K}}\mathbf{x}_{t-K-1} \end{aligned}$$

with, for $k > 0$, $\mathbf{v}_{t,k} = \phi_{s_t}\dots\phi_{s_{t-k+1}}\mathbf{v}_{t-k}$ and $\mathbf{v}_{t,0} = \mathbf{v}_t$. Let $\|\cdot\|_2$ denote the L^2-norm ($\|X\|_2 = (E(X^2))^{1/2}$). Let

$$\mathbf{x}_t^* = \sum_{k=0}^{\infty} \mathbf{v}_{t,k}$$

provided that this series converges in L^2. We have, by conditioning,

$$\|\mathbf{v}_{t,k}\|_2^2 = E(\phi_{s_t}^2\dots\phi_{s_{t-k+1}}^2\mathbf{v}_{t-k}^2) = (1,\dots,1)(\hat{\mathbf{\Phi}}^2\mathbf{P})^k \begin{pmatrix} \pi(1)(\lambda_1 + \alpha_1^2) \\ \vdots \\ \pi(K)(\lambda_K + \alpha_K^2) \end{pmatrix},$$

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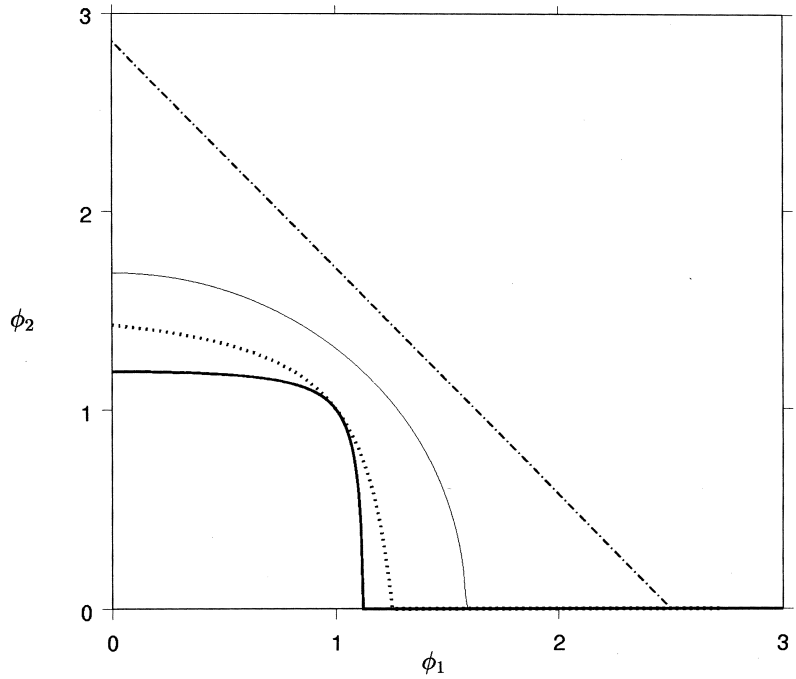


FIGURE 1. Stationarity region for the two-regime AR(1) model with $p(1,1) = 0.8$ and $p(2,2) = 0.7$. The first and second equations in (2) correspond to the thick and thin full lines, respectively. Replacing the squares by absolute values in (2) yields the dotted and dashed lines.

where the $\pi(i)$'s are the stationary probabilities of the Markov chain. So a sufficient condition for the convergence of $\sum_{k=0}^{\infty} \|\mathbf{v}_{t,k}\|_2$ is that $\mathbf{r}(\hat{\mathbf{\Phi}}^2\mathbf{P}) < 1$. By the Cauchy criterion, this condition insures the existence of $\mathbf{x}_t^*$ in L^2. Finally it is clear that the process $(\mathbf{x}_t^*)$ is a solution to (1).

Hence, in the univariate case, Assumption 2 takes the form (2).

3. In the multivariate case ($n > 1$) Assumption 2 is too restrictive. A computation similar to the previous one shows that a sufficient stationarity condition is $\mathbf{r}(\mathbf{A}) < 1$ with

$$\mathbf{A} = \begin{pmatrix} p(1,1)\{\phi_1 \otimes \phi_1\} & p(2,1)\{\phi_1 \otimes \phi_1\} & \dots & p(K,1)\{\phi_1 \otimes \phi_1\} \\ p(1,2)\{\phi_2 \otimes \phi_2\} & p(2,2)\{\phi_2 \otimes \phi_2\} & \dots & p(K,2)\{\phi_2 \otimes \phi_2\} \\ \vdots & \vdots & & \vdots \\ p(1,K)\{\phi_K \otimes \phi_K\} & p(2,K)\{\phi_K \otimes \phi_K\} & \dots & p(K,K)\{\phi_K \otimes \phi_K\} \end{pmatrix},$$

where $\otimes$ denotes the Kronecker product. So we suggest replacing Assumption 2 by the following assumption.

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Assumption 2′. $\mathbf{r}(\mathbf{A}) < 1$.

In the univariate case we have $\mathbf{A} = \hat{\mathbf{\Phi}}^2\mathbf{P}$, but this does not hold when $n > 1$. The difference between the stationarity regions implied by the two assumptions can be important. As an example, consider the two-regime AR(2) process defined by

$$X_t = \begin{cases} \eta_t & \text{if } s_t = 1 \\ \eta_t + \phi X_{t-2} & \text{if } s_t = 2, \end{cases}$$

where ϕ is a real constant and (η_t) is a white noise. This model can be written in vectorial form as in (1) with $\mathbf{x}_t = (X_t, X_{t-1})'$, $\mathbf{u}_t = (\eta_t, 0)'$, $\alpha_{s_t} = 0$, and $\phi_{s_t} = \begin{pmatrix} 0 & \phi(s_t - 1) \\ 1 & 0 \end{pmatrix}$. Straightforward computations show that Assumption 2 takes the form

$$|\phi| < 1,$$

whereas Assumption 2′ takes the form

$$|\phi| \max\{p(2,2)^{1/2}, [1 - p(2,1)(p(1,1) + p(2,2))]^{1/2}\} < 1,$$

and the max is of course less than 1. Take for illustrative purpose $p(2,1) = p(1,1) = 0.95$. Then Assumption 2′ gives $|\phi| < 4.47$.

It can be seen in this example that the stationarity condition of the paper amounts to imposing stationarity within each regime. The fact that it is too strong is quite intuitive: the coefficient $|\phi|$ can be much larger than 1 if the probability of visiting the regime 2 is small.

To conclude, let us mention that in the case of Markov-switching autoregressive moving average (ARMA) models, we have given a stationarity condition that reduces to Assumption 2′ in the present situation (see Francq and Zakoïan, 2001). This condition is however much more difficult to establish in the general ARMA case. Karlsen (1990) also obtains stationarity conditions in a general framework of time-varying AR models, whereas Poskitt and Chung (1996) and Krolzig (1997) concentrate on VAR models with Markov switching in the intercept or mean term.

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